

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester VI Examination, 2013-14

Paper No. BPT-602: Solid State Physics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer all questions. Symbols have their usual meanings.***Question 1.** Answer any **five** of the following questions: $[5 \times 4 = 20]$

- (a) With the help of diagrams, show the crystal structure of NaCl and diamond. What is the lattice type of these structures?
- (b) For metals, establish the Wiedemann-Franz law.
- (c) Show the variation of energy, velocity, and effective mass of an electron as a function of wave vector k in the first Brillouin zone.
- (d) What are the characteristic differences between the optical and acoustical modes of lattice vibrations?
- (e) What do you understand by the Lennard-Jones potential?
- (f) Explain the meaning of spin waves and magnons.

Question 2.

- (a) What is the meaning of a space-lattice? How many different types of space-lattices are possible? $[6]$
- (b) Prove that the close packing of atoms in the hexagonal close-packed (HCP) structure demands an axial ratio $c/a = \sqrt{8/3} \approx 1.633$. $[6]$

Question 3.

- (a) Using the Einstein model, derive the expression for the specific heat of solids at high and low temperatures. What are the assumptions of the Debye model of specific heat of solids? How does it differ from the Einstein model? $[8]$
- (b) Considering the nearest neighbour interaction, establish the dispersion relation for a monoatomic linear chain. $[5]$

OR

- (a) Derive the expression for the energy of conduction electrons in a metal considering its potential energy to be zero. $[8]$
- (b) On the basis of the free electron gas model, calculate the expression for electrical conductivity. $[5]$

Question 4.

- (a) Discuss the quantum theory of paramagnetism and establish the Curie-Brillouin law for the magnetization and hence the Curie law of susceptibility. [8]
- (b) What is the Weiss field for ferromagnetism? Establish the Curie-Weiss law for ferromagnetism. [5]

OR

- (a) What do you understand by reciprocal lattice? [2]
- (b) Show that the reciprocal vector $\vec{G}_{hkl}$ is perpendicular to the (hkl) plane and $|\vec{G}_{hkl}| = \frac{2\pi}{d_{hkl}}$. [6]
- (c) What is the Ewald construction? [5]

Question 5.

- (a) Derive the expressions for the atomic scattering factor and crystal structure factor. [8]
- (b) Calculate the structure factor for a body-centered cubic (BCC) structure and show that its value is non-zero only when $h + k + l$ is an even number, where h , k , and l are the Miller indices of the plane. [4]